FINAL PROJECT

FortiGate Firewall Security Configuration Lab

Network Access Control, Web Filtering, and Application Control Implementation

1. Introduction

This project demonstrates the configuration and implementation of a FortiGate firewall in a virtualized lab environment to enforce network security policies. The objective was to design and implement firewall rules that allow controlled internet access while enforcing security restrictions using Web Filtering and Application Control.

The lab environment consisted of two Ubuntu virtual machines (PC-1 and PC-2) connected through a FortiGate firewall. Security policies were configured to permit general internet connectivity while selectively restricting specific websites and application categories.

This project highlights practical skills in firewall configuration, traffic filtering, and security policy enforcement using Fortinet technologies.

2. Lab Environment

Virtual Machines and Devices

The following devices were used in the lab setup:

Device	Interface	IP Address	Purpose
FortiGate Firewall	Port1	DHCP	WAN / Internet access
FortiGate Firewall	Port2	10.10.10.1	Internal LAN Gateway
PC-1	ens33	10.10.10.10	Internal client
PC-2	ens33	10.10.10.11	Internal client

Both client machines were connected to the same LAN segment through the FortiGate firewall.

3. Network Architecture

PC-1 (10.10.10.10)

|

PC-2 (10.10.10.11)

|

|

FortiGate Firewall (10.10.10.1)

|

|

Internet

The FortiGate firewall served as the central security gateway controlling all inbound and outbound traffic.

4. Task 1 – Full Internet Access Rule

Objective

Allow both internal hosts (PC-1 and PC-2) to access the internet.

Configuration

A firewall policy was created allowing outbound traffic from the internal network (Port2) to the WAN interface (Port1).

Policy Configuration

Parameter	Value
Incoming Interface	Port2
Outgoing Interface	Port1
Source	PC-1 & PC-2
Destination	All

Parameter	Value
Service	ALL
Schedule	Always
NAT	Enabled

Verification

Connectivity was confirmed by successfully pinging Google's public DNS server.

Example verification command:

ping 8.8.8.8

Both PC-1 and PC-2 successfully accessed external networks, confirming that the firewall rule allowed outbound internet connectivity.

5. Task 2 – Web Filtering (PC-1)

Objective

Block access to specific websites from PC-1.

Target Websites

- williscollege.com
- willisonline.ca

Configuration

A **Web Filter profile** was created and applied to a firewall policy specific to PC-1.

Web Filter Settings

Setting	Action
williscollege.com	Block
willisonline.ca	Block

The web filter profile was then applied to the firewall policy governing traffic from PC-1.

Result

Attempts to access the blocked websites from PC-1 resulted in blocked access events recorded in the FortiGate security logs.

6. Task 3 – Application Control (PC-2)

Objective

Restrict video streaming applications while allowing normal web access.

Configuration

An **Application Control profile** was created to block applications categorized under **Video/Audio**.

Application Control Profile

Category	Action
Video/Audio	Block

This profile was applied to the firewall policy controlling traffic from PC-2.

7. Verification of Application Control

Allowed Traffic

Normal internet services such as DNS queries and standard web browsing remained accessible.

Blocked Traffic

Attempts to access video streaming platforms such as YouTube resulted in connection timeouts, indicating successful blocking by the firewall.

Example blocked site:

youtube.com

Security event logs within the FortiGate interface confirmed that the application was blocked by the configured Application Control profile.

8. Security Logs Verification

The FortiGate **Application Control Logs** were used to verify policy enforcement.

Logs displayed entries indicating that YouTube traffic originating from PC-2 was blocked according to the configured application control rule.

This confirms that the firewall successfully identified and restricted video streaming traffic.

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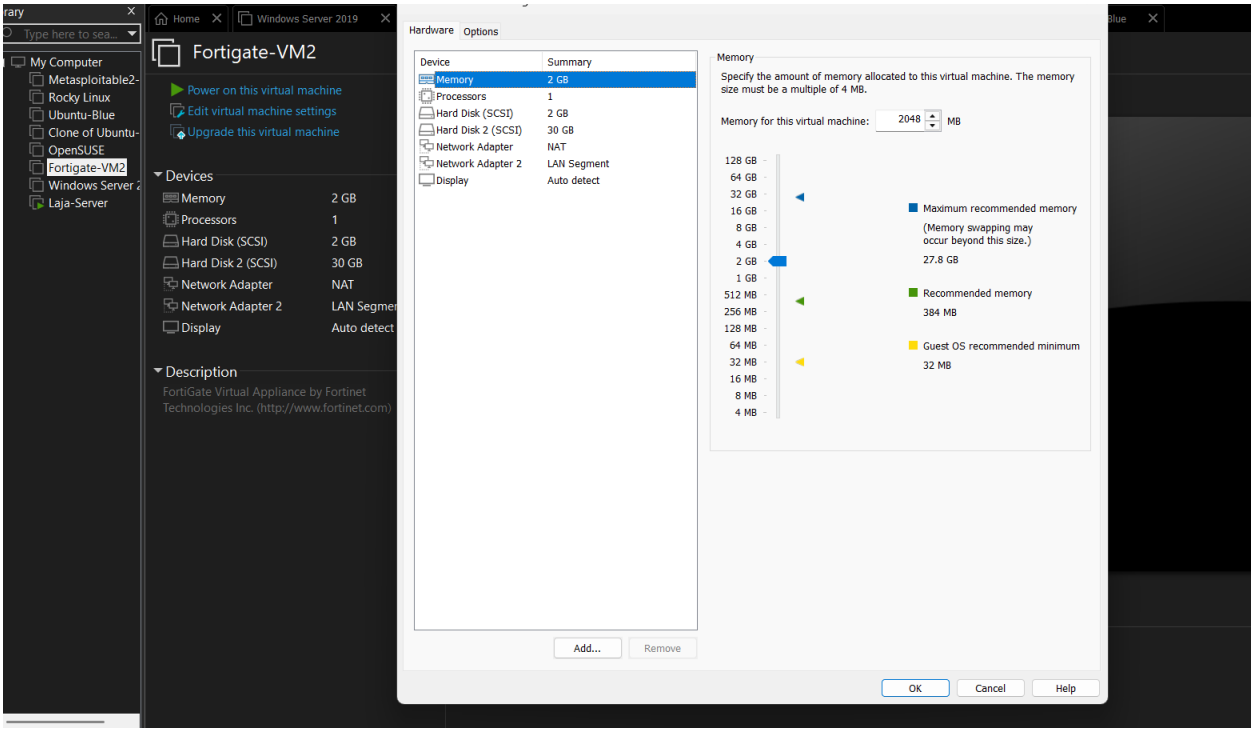
9. Results Summary

Security Control	Result
Internet Access	Allowed for both PCs
Web Filtering	Willis College sites blocked for PC-1
Application Control	Video streaming blocked for PC-2
Logging	Security events successfully recorded

The firewall policies effectively controlled network access while maintaining visibility through monitoring logs.

TASK 1:

Attach Fortigate Adapter:



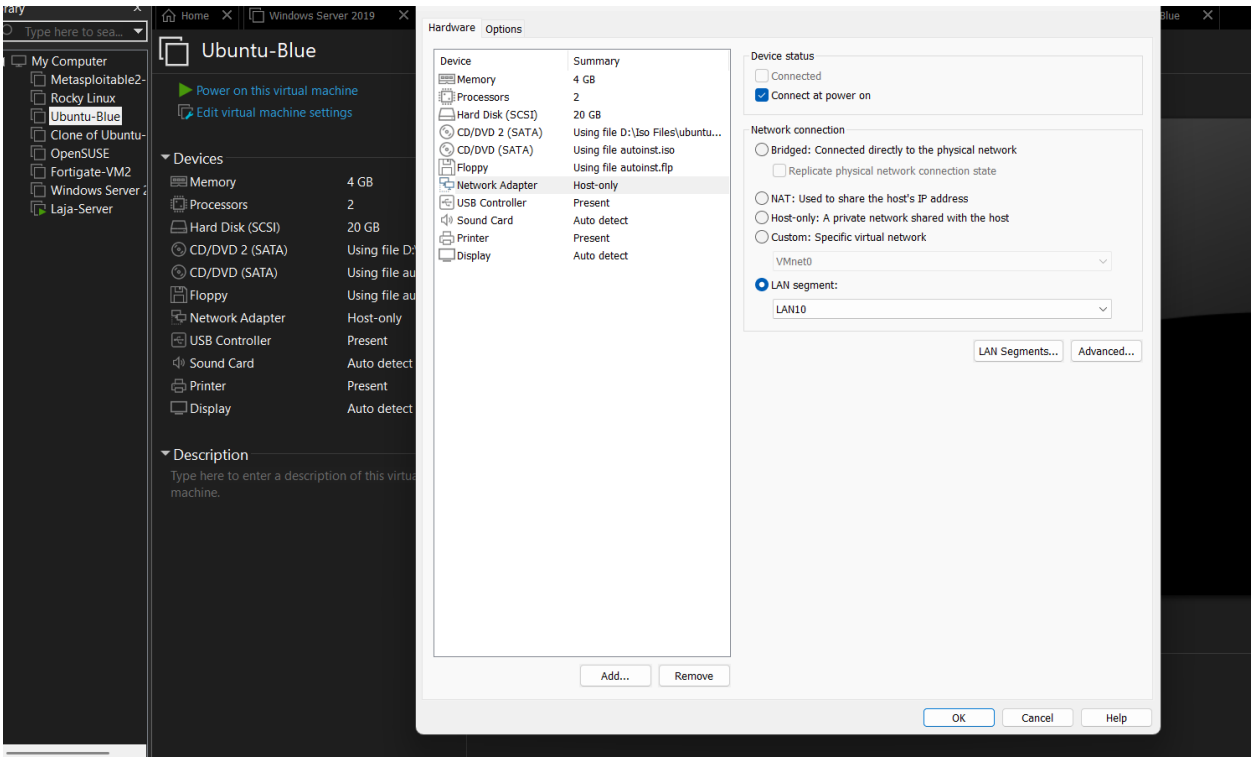
Show Available interfaces configured:

```
end

FGVMEV-ZSYLBHF23 # show system interface physical
entry is not found in table

FGVMEV-ZSYLBHF23 # get system interface physical
== [onboard]
        ==[port1]
                mode: dhcp
                ip: 192.168.42.140 255.255.255.0
                ipv6: ::/0
                status: up
                speed: 10000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none
        ==[port2]
                mode: static
                ip: 10.10.10.1 255.255.255.0
                ipv6: ::/0
                status: up
                speed: 10000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none

FGVMEV-ZSYLBHF23 #
```

Configure port 1 and port 2:

```
FGVMEV-ZSYLBHF23 (interface) # edit port2

FGVMEV-ZSYLBHF23 (port2) # set mode static

FGVMEV-ZSYLBHF23 (port2) # set ip 10.10.10.1 255.255.255.0

FGVMEV-ZSYLBHF23 (port2) # set allowaccess ping https ssh http

FGVMEV-ZSYLBHF23 (port2) # next

FGVMEV-ZSYLBHF23 (interface) # end

FGVMEV-ZSYLBHF23 # show system interface port2
config system interface
    edit "port2"
        set vdom "root"
        set ip 10.10.10.1 255.255.255.0
        set allowaccess ping https ssh http
        set type physical
        set alias "WAN"
        set snmp-index 2
    next
end

FGVMEV-ZSYLBHF23 # _
```

Attach PC1 and PC2(Ubuntu-Blue + Cloned)) TO LAN10:



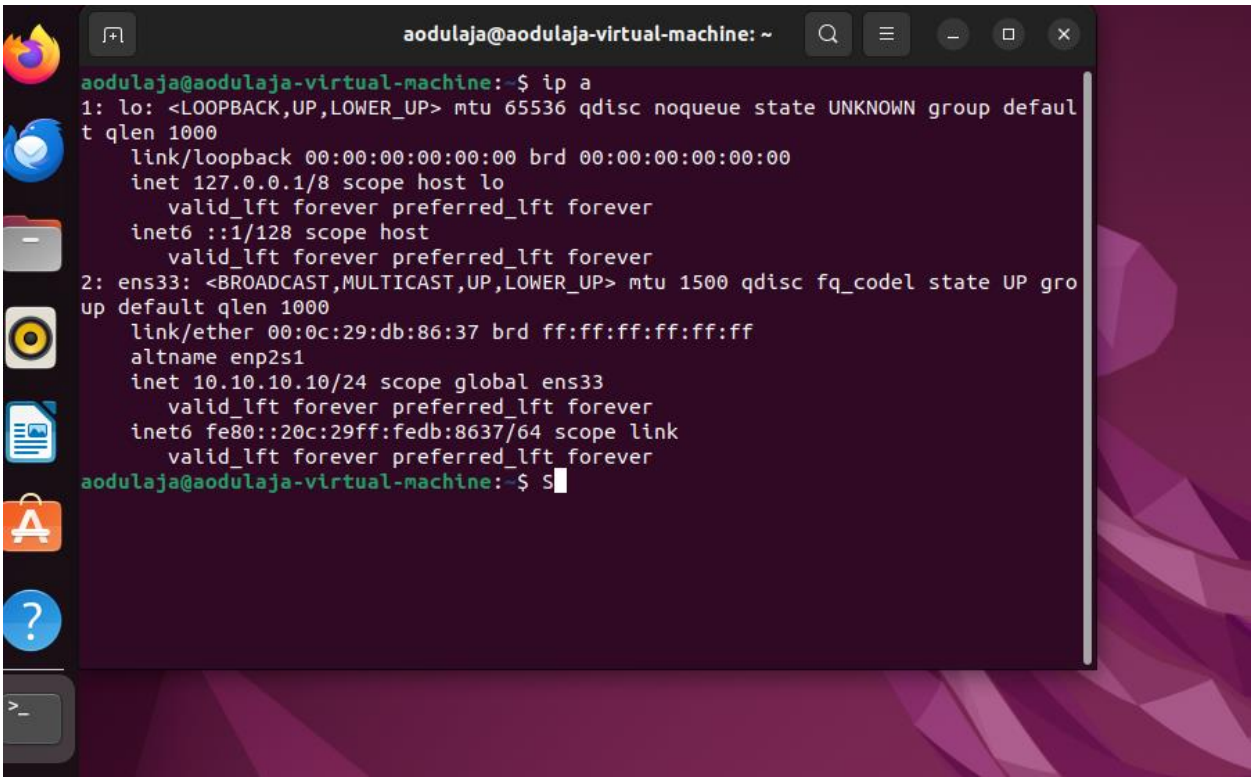
Verify Routing Table:

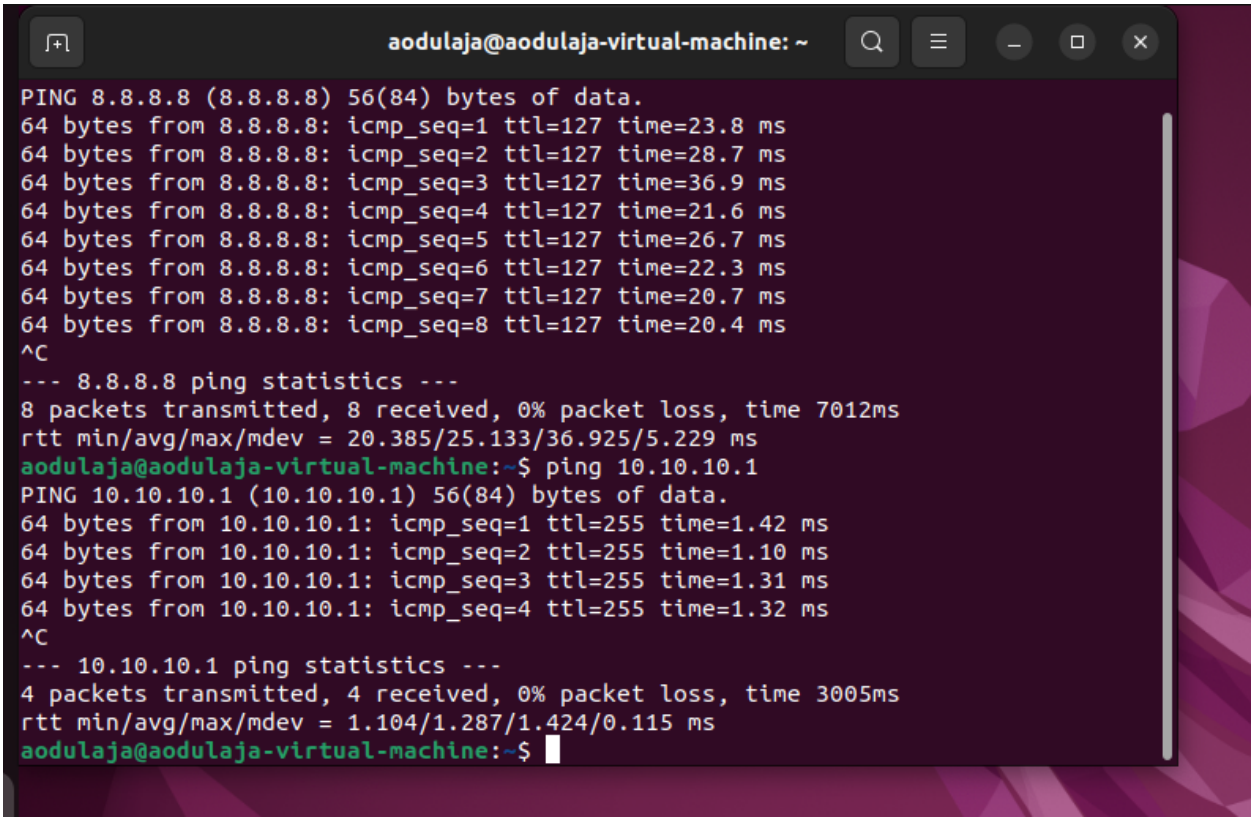
```
        ==[port2]
                mode: static
                ip: 10.10.10.1 255.255.255.0
                ipv6: ::/0
                status: up
                speed: 10000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none

FGVMEV-ZSYLBHF23 # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
S*      0.0.0.0/0 [5/0] via 192.168.42.2, port1, [1/0]
C       10.10.10.0/24 is directly connected, port2
C       192.168.42.0/24 is directly connected, port1


FGVMEV-ZSYLBHF23 #
```

Set Firewall Policy:

```
new entry '1' added
FGVMEV-ZSYLBHF23 (1) # set name "LAN_to_WAN"
FGVMEV-ZSYLBHF23 (1) # set srcintf port2
FGVMEV-ZSYLBHF23 (1) # set dstintf port1
FGVMEV-ZSYLBHF23 (1) # set srcaddr all
FGVMEV-ZSYLBHF23 (1) # set dstaddr all
FGVMEV-ZSYLBHF23 (1) # set action accept
FGVMEV-ZSYLBHF23 (1) # set schedule always
FGVMEV-ZSYLBHF23 (1) # set service ALL
FGVMEV-ZSYLBHF23 (1) # set nat enable
FGVMEV-ZSYLBHF23 (1) # next
FGVMEV-ZSYLBHF23 (policy) # end
FGVMEV-ZSYLBHF23 #
```

TEST POLICY FROM PC1:





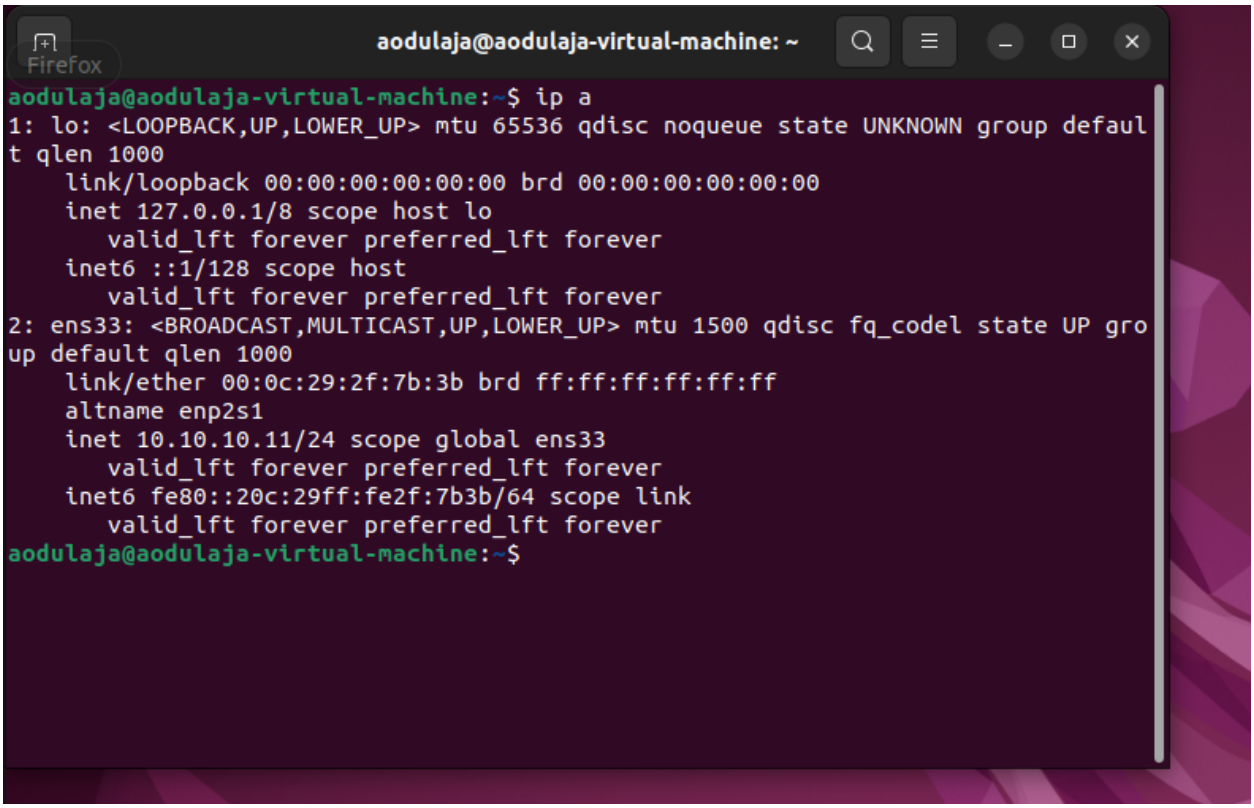
aodulaja@aodulaja-virtual-machine: ~

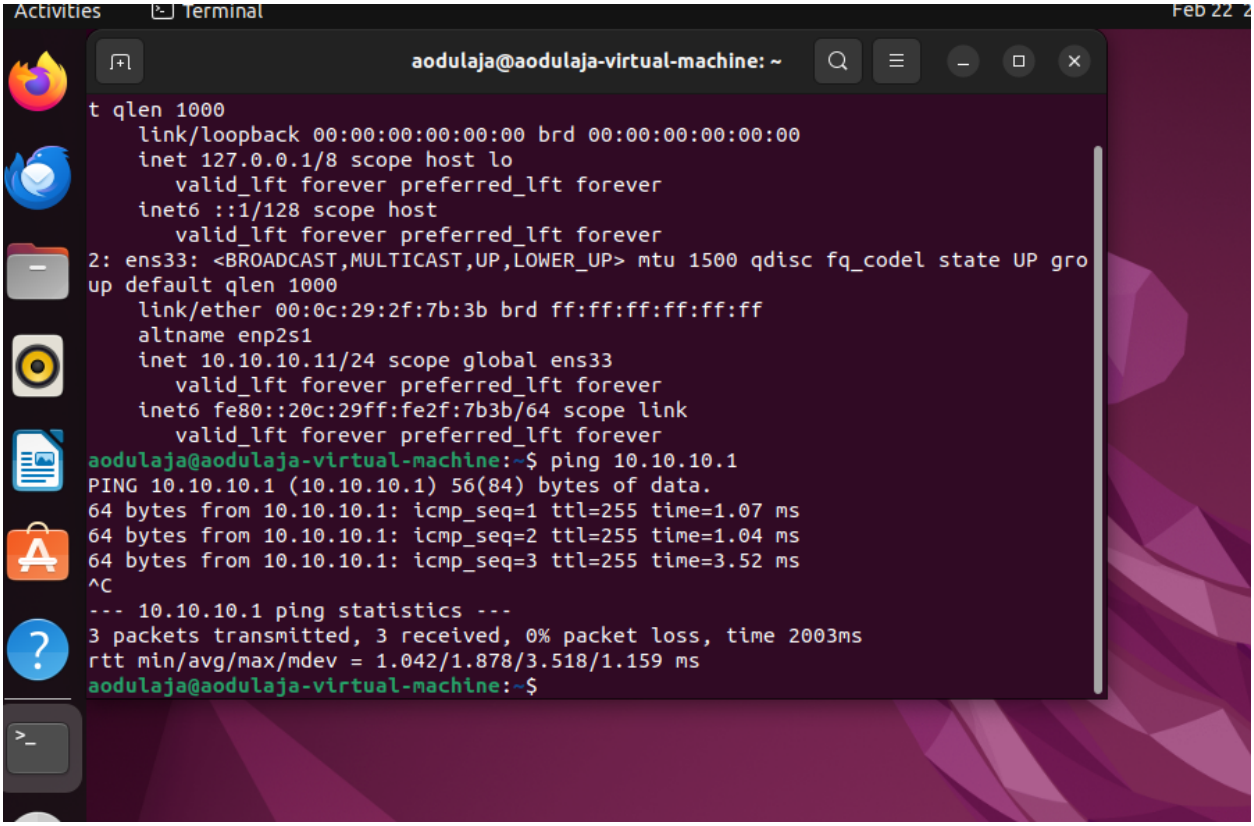
```
--- 8.8.8.8 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7012ms
rtt min/avg/max/mdev = 20.385/25.133/36.925/5.229 ms
aodulaja@aodulaja-virtual-machine:~$ ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=255 time=1.42 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=255 time=1.10 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=255 time=1.31 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=255 time=1.32 ms
^C
--- 10.10.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.104/1.287/1.424/0.115 ms
aodulaja@aodulaja-virtual-machine:~$ ping 10.10.10.11
PING 10.10.10.11 (10.10.10.11) 56(84) bytes of data.
64 bytes from 10.10.10.11: icmp_seq=1 ttl=64 time=2.34 ms
64 bytes from 10.10.10.11: icmp_seq=2 ttl=64 time=1.17 ms
64 bytes from 10.10.10.11: icmp_seq=3 ttl=64 time=1.21 ms
64 bytes from 10.10.10.11: icmp_seq=4 ttl=64 time=1.07 ms
^C
--- 10.10.10.11 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.066/1.445/2.335/0.516 ms
aodulaja@aodulaja-virtual-machine:~$
```

aodulaja@aodulaja-virtual-machine: ~

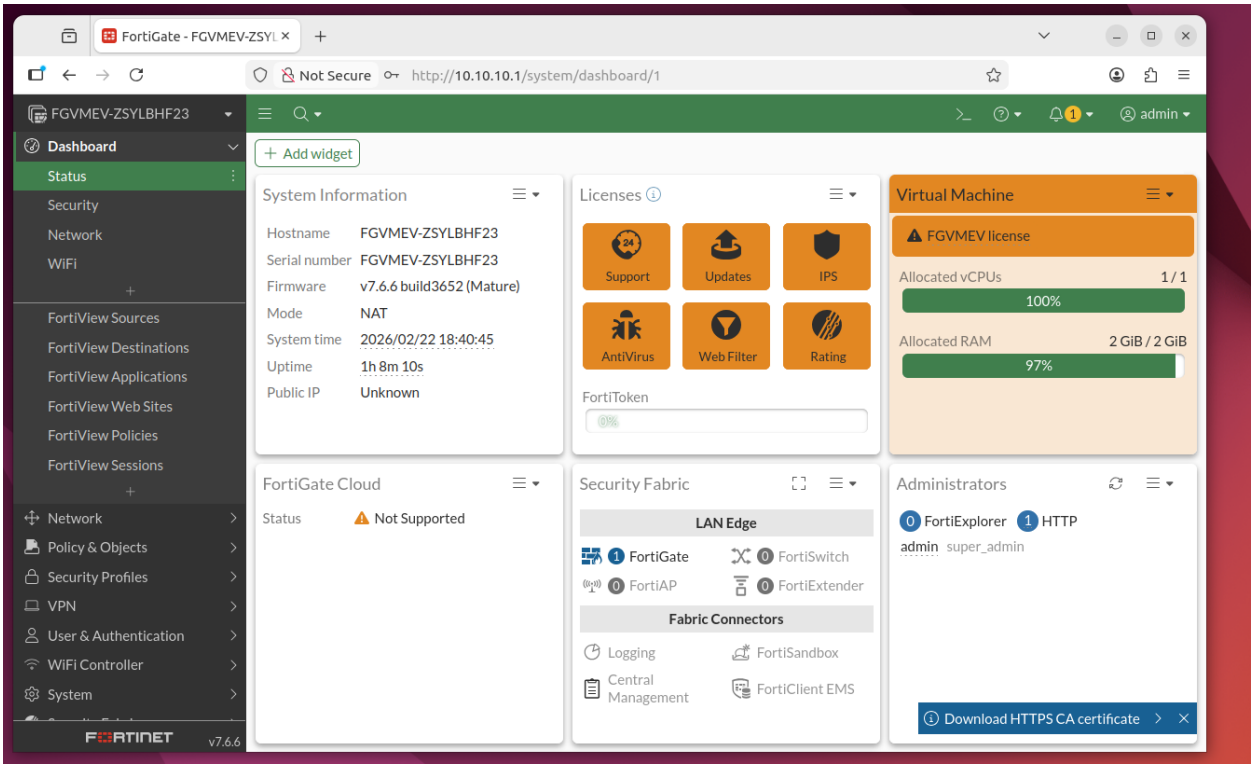
```
64 bytes from 10.10.10.11: icmp_seq=4 ttl=64 time=1.07 ms
^C
--- 10.10.10.11 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.066/1.445/2.335/0.516 ms
aodulaja@aodulaja-virtual-machine:~$ ping google.com
ping: google.com: Temporary failure in name resolution
aodulaja@aodulaja-virtual-machine:~$ sudo nano /etc/resolv.conf
[sudo] password for aodulaja:
aodulaja@aodulaja-virtual-machine:~$ ping google.com
PING google.com (142.250.69.110) 56(84) bytes of data.
64 bytes from pnyula-ab-in-f14.1e100.net (142.250.69.110): icmp_seq=1 ttl=127 ti
me=22.2 ms
64 bytes from pnyula-ab-in-f14.1e100.net (142.250.69.110): icmp_seq=2 ttl=127 ti
me=19.5 ms
64 bytes from pnyula-ab-in-f14.1e100.net (142.250.69.110): icmp_seq=3 ttl=127 ti
me=22.7 ms
64 bytes from pnyula-ab-in-f14.1e100.net (142.250.69.110): icmp_seq=4 ttl=127 ti
me=29.0 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3000ms
rtt min/avg/max/mdev = 19.456/23.364/29.040/3.506 ms
aodulaja@aodulaja-virtual-machine:~$
```

FROM PC2:

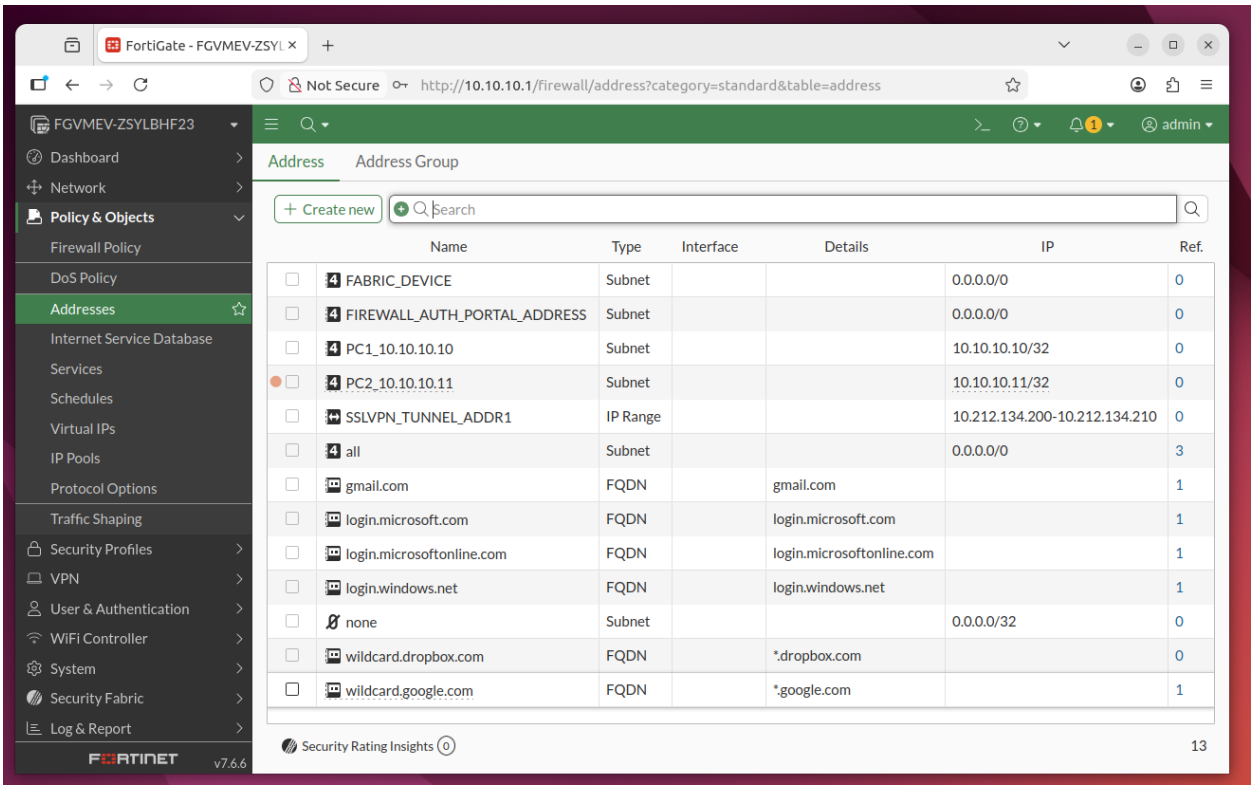

aodulaja@aodulaja-virtual-machine: ~
aodulaja@aodulaja-virtual-machine:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
link/ether 00:0c:29:2f:7b:3b brd ff:ff:ff:ff:ff:ff
altname enp2s1
inet 10.10.10.11/24 scope global ens33
valid_lft forever preferred_lft forever
inet6 fe80::20c:29ff:fe2f:7b3b/64 scope link
valid_lft forever preferred_lft forever
aodulaja@aodulaja-virtual-machine:~$


Activities
Terminal
aodulaja@aodulaja-virtual-machine: ~
t qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
link/ether 00:0c:29:2f:7b:3b brd ff:ff:ff:ff:ff:ff
altname enp2s1
inet 10.10.10.11/24 scope global ens33
valid_lft forever preferred_lft forever
inet6 fe80::20c:29ff:fe2f:7b3b/64 scope link
valid_lft forever preferred_lft forever
aodulaja@aodulaja-virtual-machine:~$ ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=255 time=1.07 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=255 time=1.04 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=255 time=3.52 ms
^C
--- 10.10.10.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 1.042/1.878/3.518/1.159 ms
aodulaja@aodulaja-virtual-machine:~$

TASK 2:

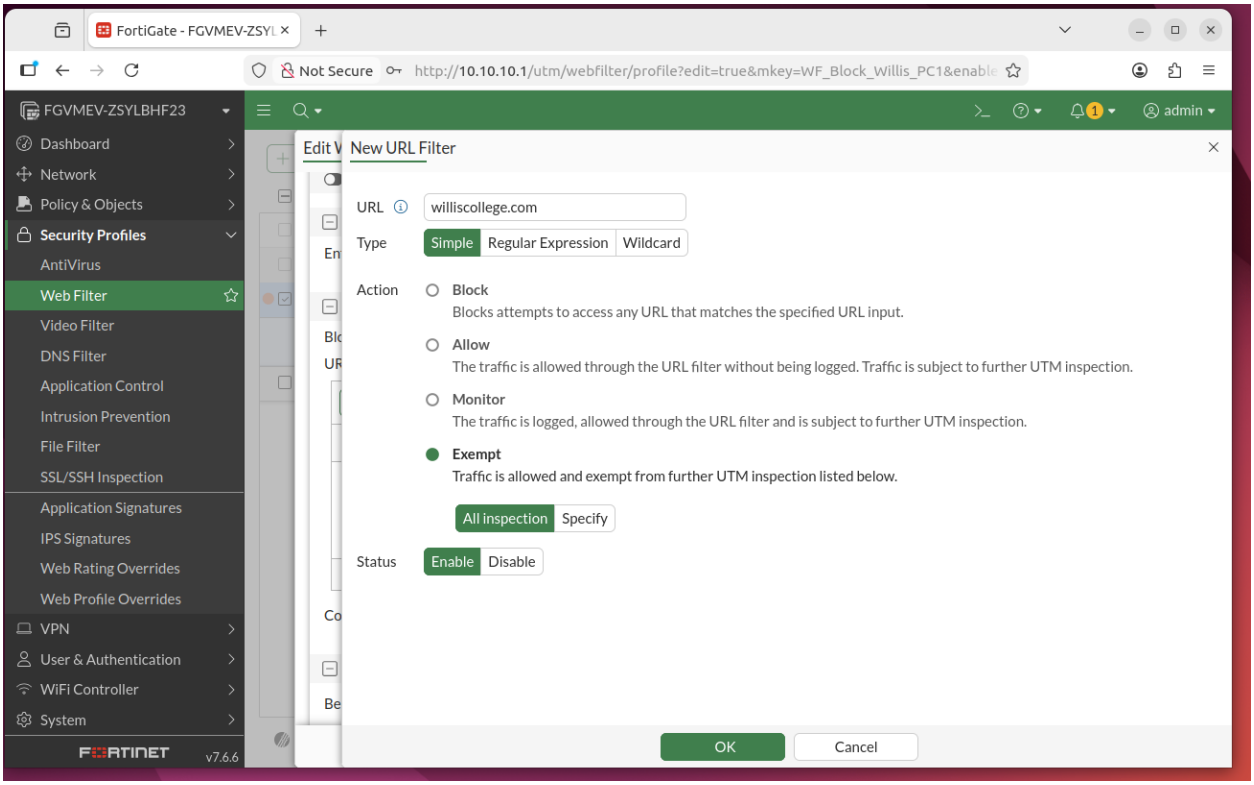

FortiGate - FGVMEV-ZSYL
Not Secure
http://10.10.10.1/system/dashboard/1
FGVMEV-ZSYLBHF23
admin
Dashboard
Status
Security
Network
WiFi
FortiView Sources
FortiView Destinations
FortiView Applications
FortiView Web Sites
FortiView Policies
FortiView Sessions
Network
Policy & Objects
Security Profiles
VPN
User & Authentication
WiFi Controller
System
FORTINET
v7.6.6
Add widget
System Information
Hostname FGVMEV-ZSYLBHF23
Serial number FGVMEV-ZSYLBHF23
Firmware v7.6.6 build3652 (Mature)
Mode NAT
System time 2026/02/22 18:40:45
Uptime 1h 8m 10s
Public IP Unknown
Licenses
Support
Updates
IPS
AntiVirus
Web Filter
Rating
FortiToken
Virtual Machine
FGVMEV license
Allocated vCPUs 1 / 1
100%
Allocated RAM 2 GiB / 2 GiB
97%
FortiGate Cloud
Status Not Supported
Security Fabric
LAN Edge
1 FortiGate
0 FortiSwitch
0 FortiAP
0 FortiExtender
Fabric Connectors
Logging
FortiSandbox
Central Management
FortiClient EMS
Administrators
0 FortiExplorer
1 HTTP
admin super_admin
Download HTTPS CA certificate

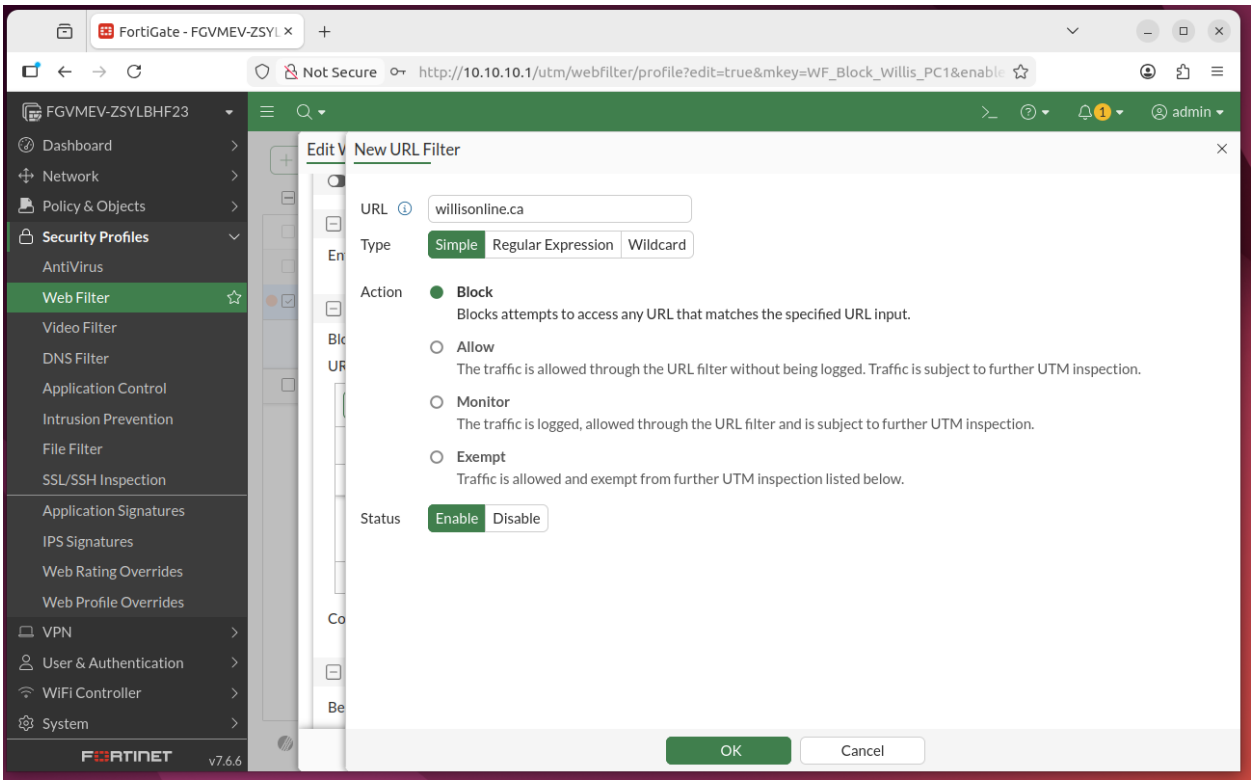
Creating Address Object for PC1 & PC2 :

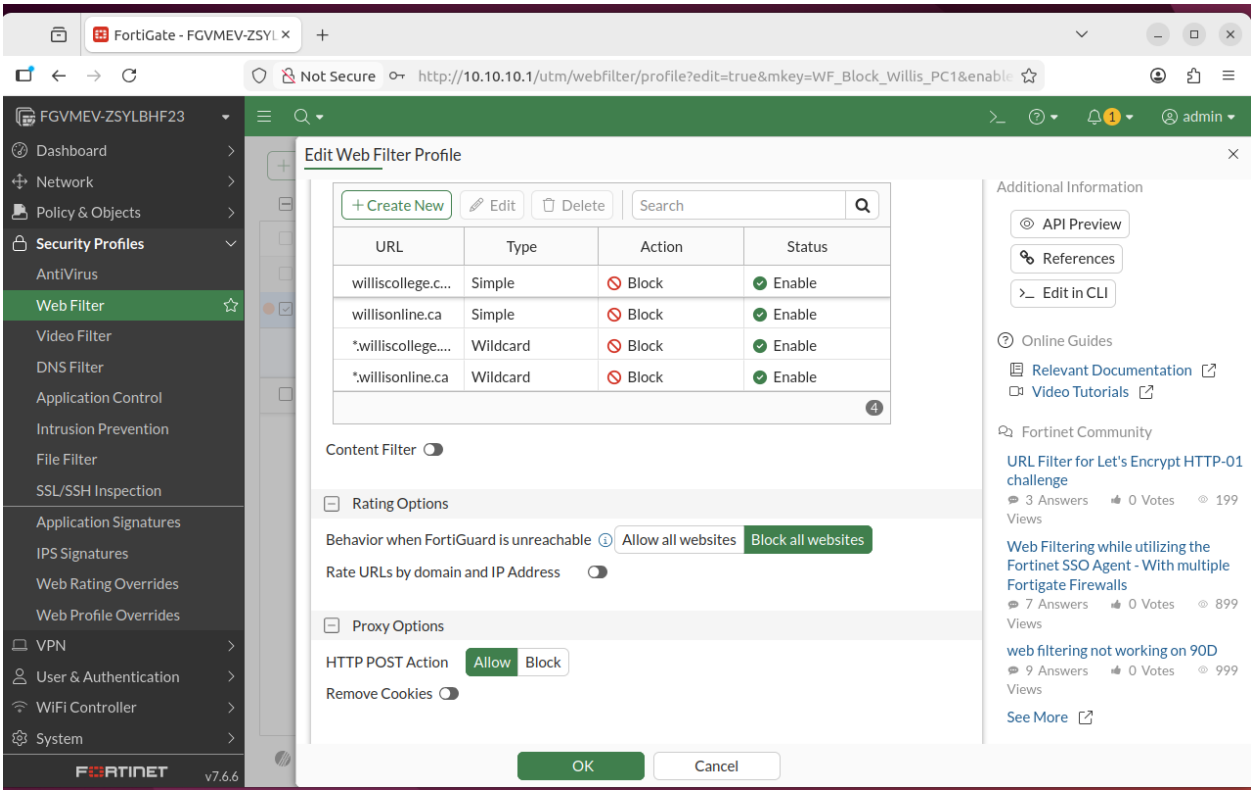


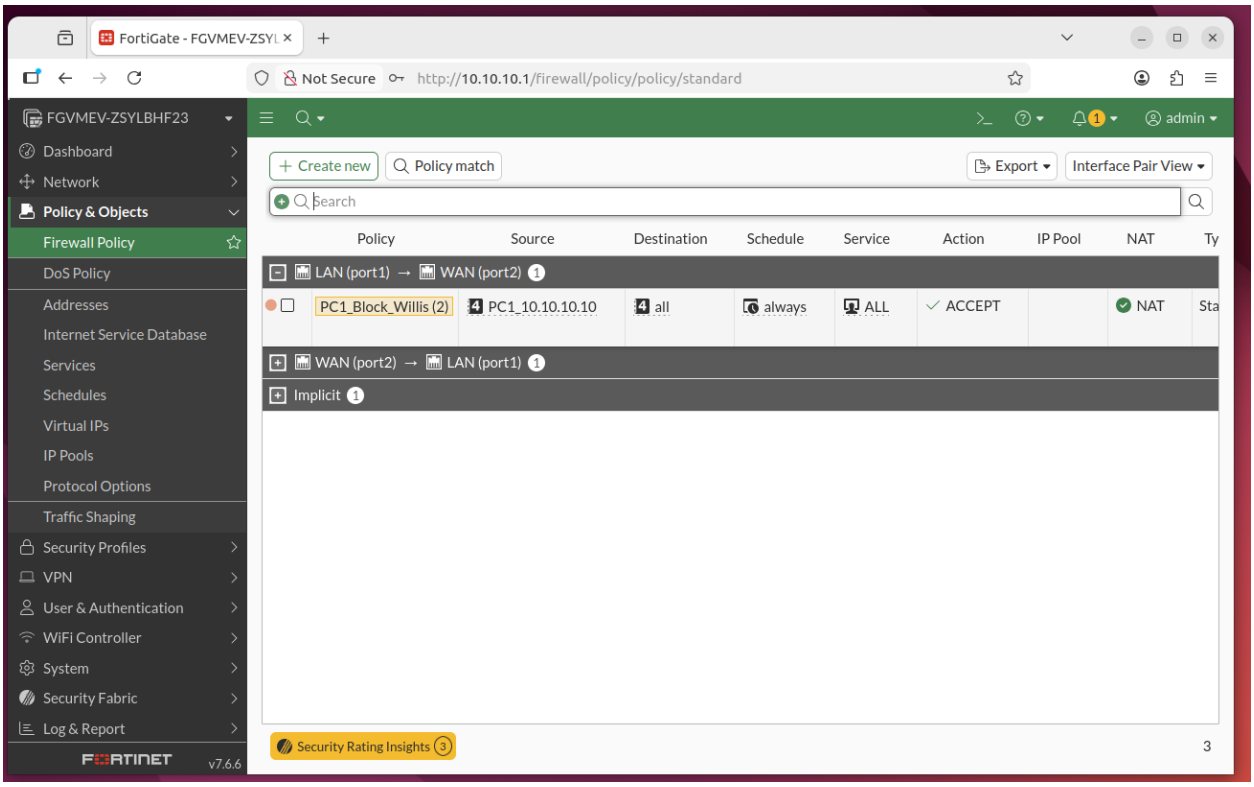
WEB FILTER

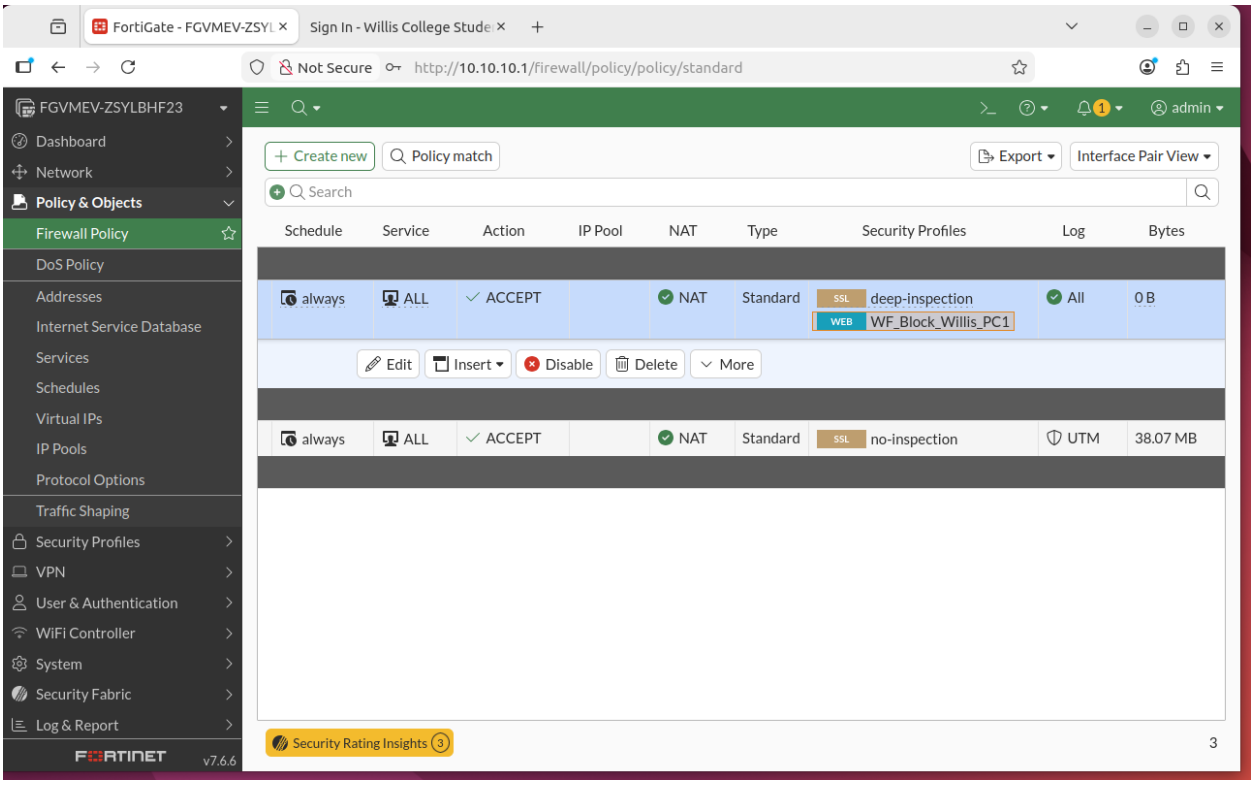
BloCK WillisCollege For PC1:

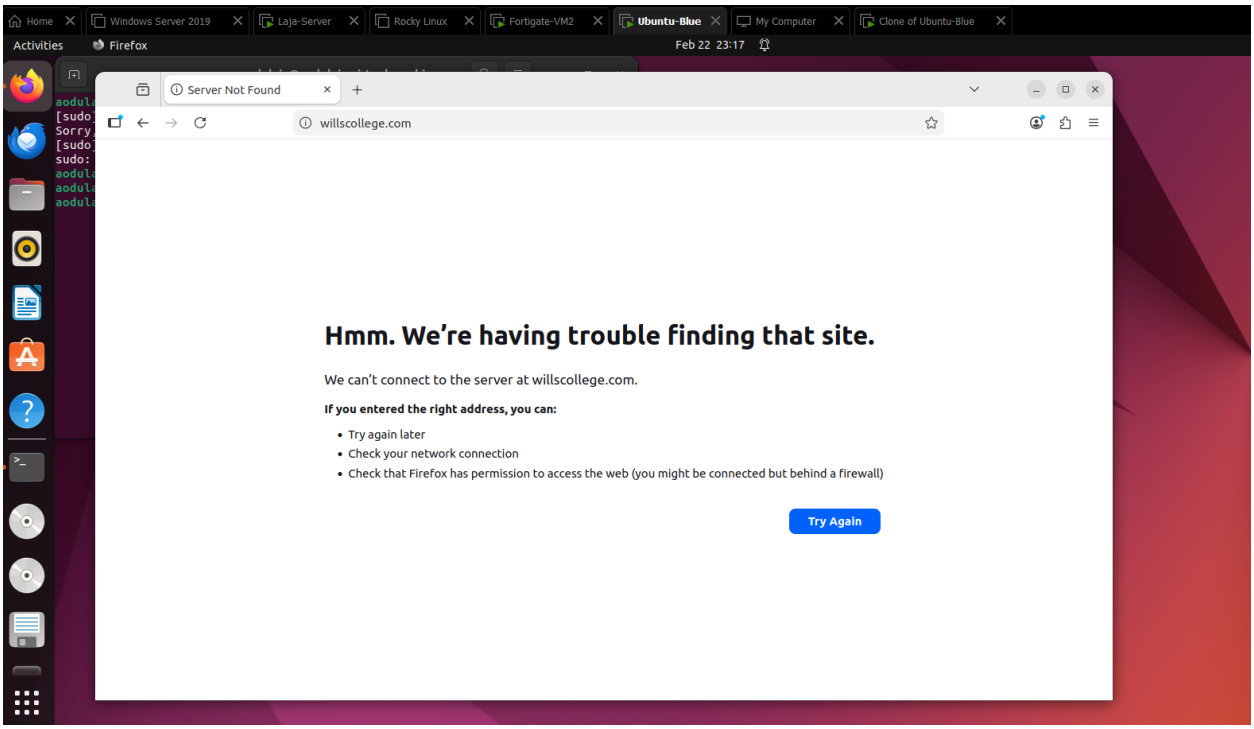





FortiGate - FGVMEV-ZSYL
Not Secure
http://10.10.10.1/utm/webfilter/profile?edit=true&mkey=WF_Block_Willis_PC1&enable
FGVMEV-ZSYLBHF23
admin
Dashboard
Network
Policy & Objects
Security Profiles
AntiVirus
Web Filter
Video Filter
DNS Filter
Application Control
Intrusion Prevention
File Filter
SSL/SSH Inspection
Application Signatures
IPS Signatures
Web Rating Overrides
Web Profile Overrides
VPN
User & Authentication
WiFi Controller
System
v7.6.6
Edit Web Filter Profile
Create New
Edit
Delete
Search
URL
Type
Action
Status
williscollege.c...
Simple
Block
Enable
willisonline.ca
Simple
Block
Enable
*.williscollege....
Wildcard
Block
Enable
*.willisonline.ca
Wildcard
Block
Enable
Content Filter
Rating Options
Behavior when FortiGuard is unreachable
Allow all websites
Block all websites
Rate URLs by domain and IP Address
Proxy Options
HTTP POST Action
Allow
Block
Remove Cookies
OK
Cancel
Additional Information
API Preview
References
Edit in CLI
Online Guides
Relevant Documentation
Video Tutorials
Fortinet Community
URL Filter for Let's Encrypt HTTP-01 challenge
3 Answers
0 Votes
199 Views
Web Filtering while utilizing the Fortinet SSO Agent - With multiple Fortigate Firewalls
7 Answers
0 Votes
899 Views
web filtering not working on 90D
9 Answers
0 Votes
999 Views
See More

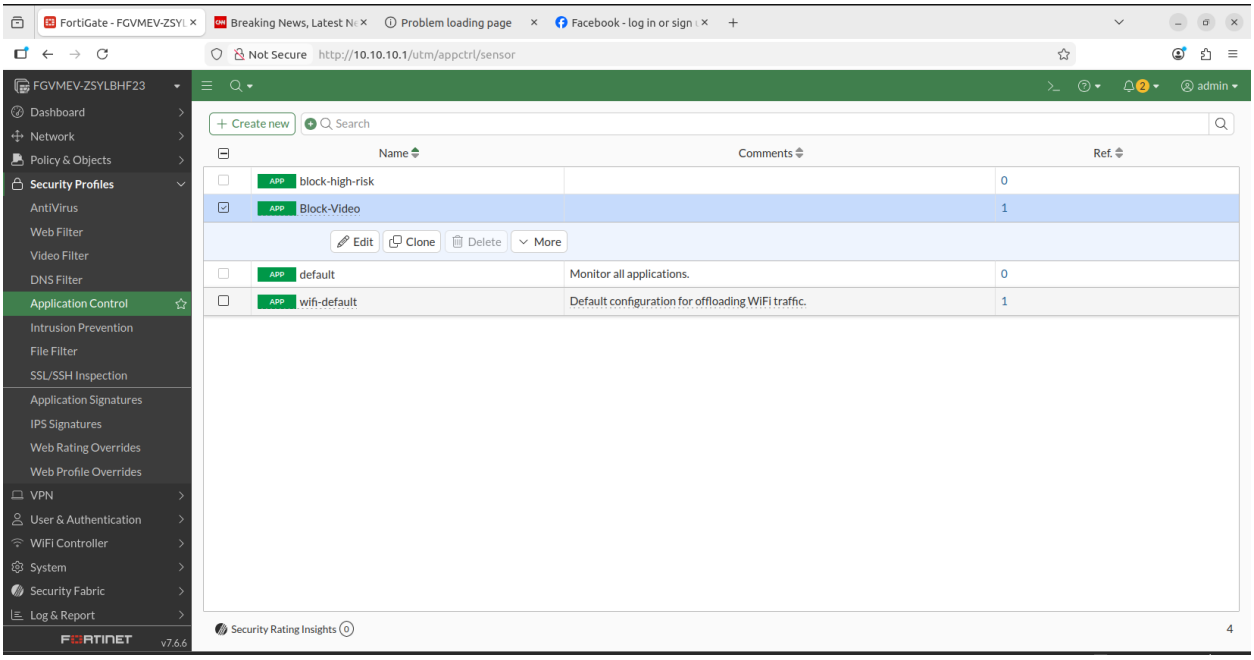

FortiGate - FGVMEV-ZSYL
Not Secure
http://10.10.10.1/firewall/policy/policy/standard
FGVMEV-ZSYLBHF23
admin
Dashboard
Network
Policy & Objects
Firewall Policy
DoS Policy
Addresses
Internet Service Database
Services
Schedules
Virtual IPs
IP Pools
Protocol Options
Traffic Shaping
Security Profiles
VPN
User & Authentication
WiFi Controller
System
Security Fabric
Log & Report
v7.6.6
Create new
Policy match
Export
Interface Pair View
Search
Policy
Source
Destination
Schedule
Service
Action
IP Pool
NAT
LAN (port1) → WAN (port2)
PC1_Block_Willis (2)
PC1_10.10.10.10
all
always
ALL
ACCEPT
NAT
WAN (port2) → LAN (port1)
Implicit
Security Rating Insights
3


FortiGate - FGVMEV-ZSYL
Sign In - Willis College Stude
Not Secure
http://10.10.10.1/firewall/policy/policy/standard
FGVMEV-ZSYLBHF23
admin
Dashboard
Network
Policy & Objects
Firewall Policy
DoS Policy
Addresses
Internet Service Database
Services
Schedules
Virtual IPs
IP Pools
Protocol Options
Traffic Shaping
Security Profiles
VPN
User & Authentication
WiFi Controller
System
Security Fabric
Log & Report
FORTINET
v7.6.6
Create new
Policy match
Export
Interface Pair View
Search
Schedule
Service
Action
IP Pool
NAT
Type
Security Profiles
Log
Bytes
always
ALL
ACCEPT
NAT
Standard
SSL
deep-inspection
WEB
WF_Block_Willis_PC1
All
0 B
Edit
Insert
Disable
Delete
More
always
ALL
ACCEPT
NAT
Standard
SSL
no-inspection
UTM
38.07 MB
Security Rating Insights
3

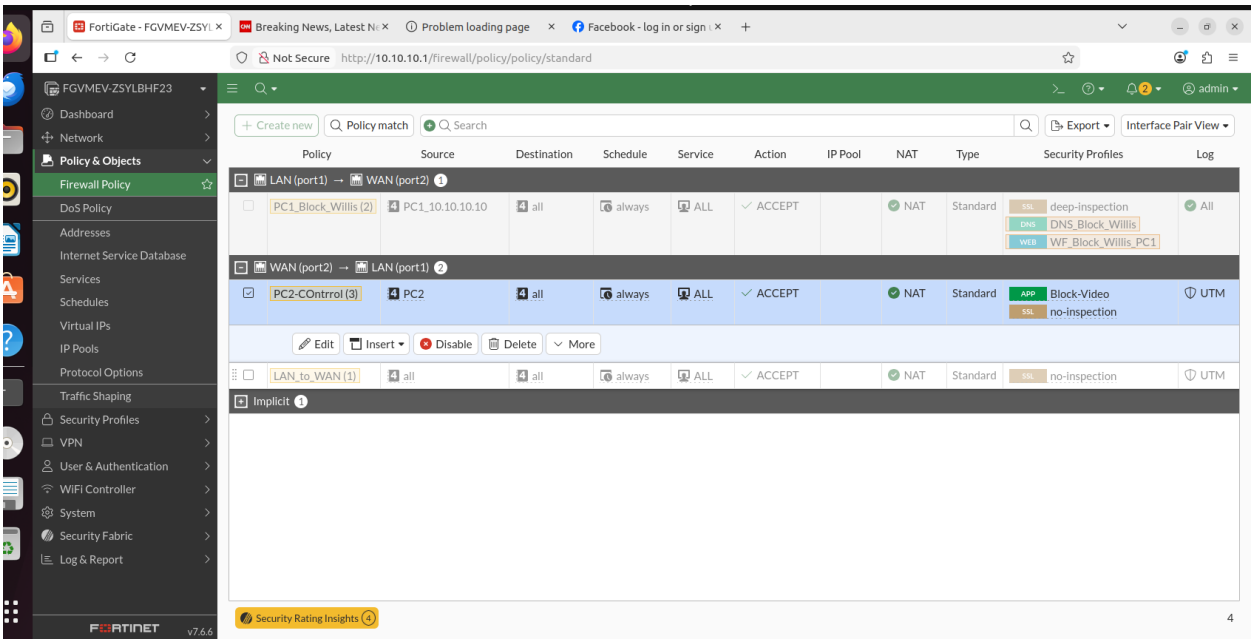

Home
Windows Server 2019
Laja-Server
Rocky Linux
Fortigate-VM2
Ubuntu-Blue
My Computer
Clone of Ubuntu-Blue
Activities
Firefox
Feb 22 23:17
Server Not Found
willscollege.com
Hmm. We're having trouble finding that site.
We can't connect to the server at willscollege.com.
If you entered the right address, you can:
Try again later
Check your network connection
Check that Firefox has permission to access the web (you might be connected but behind a firewall)
Try Again

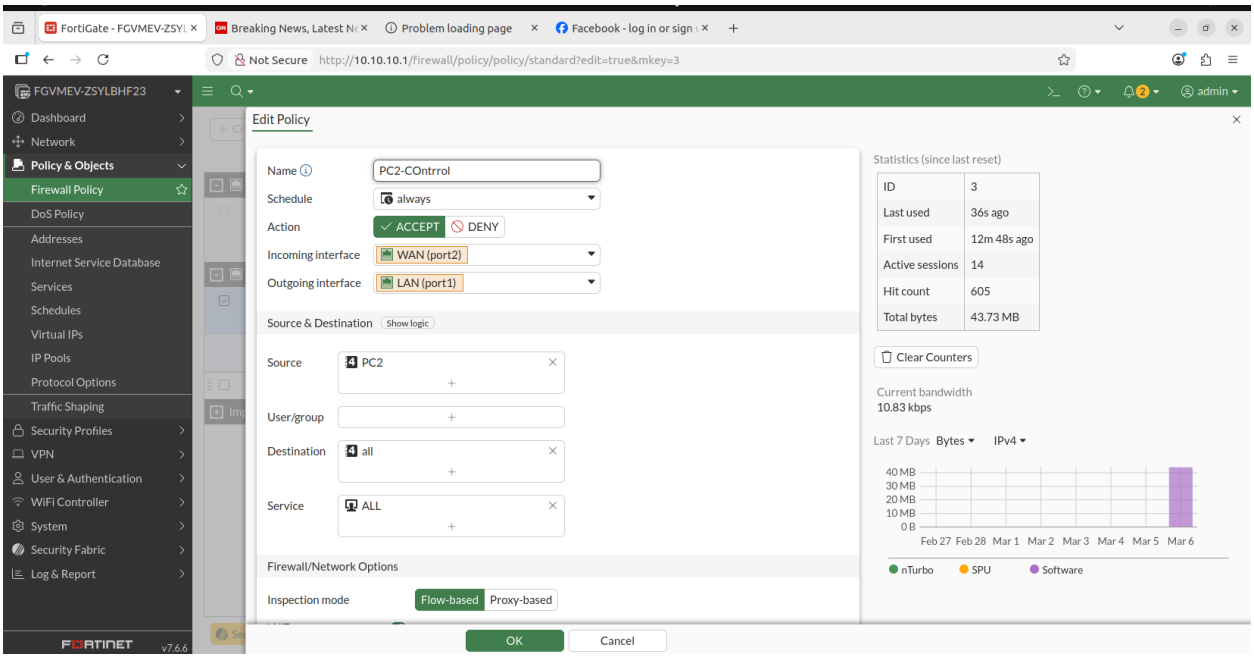
Task 3:

Application Control Profile:

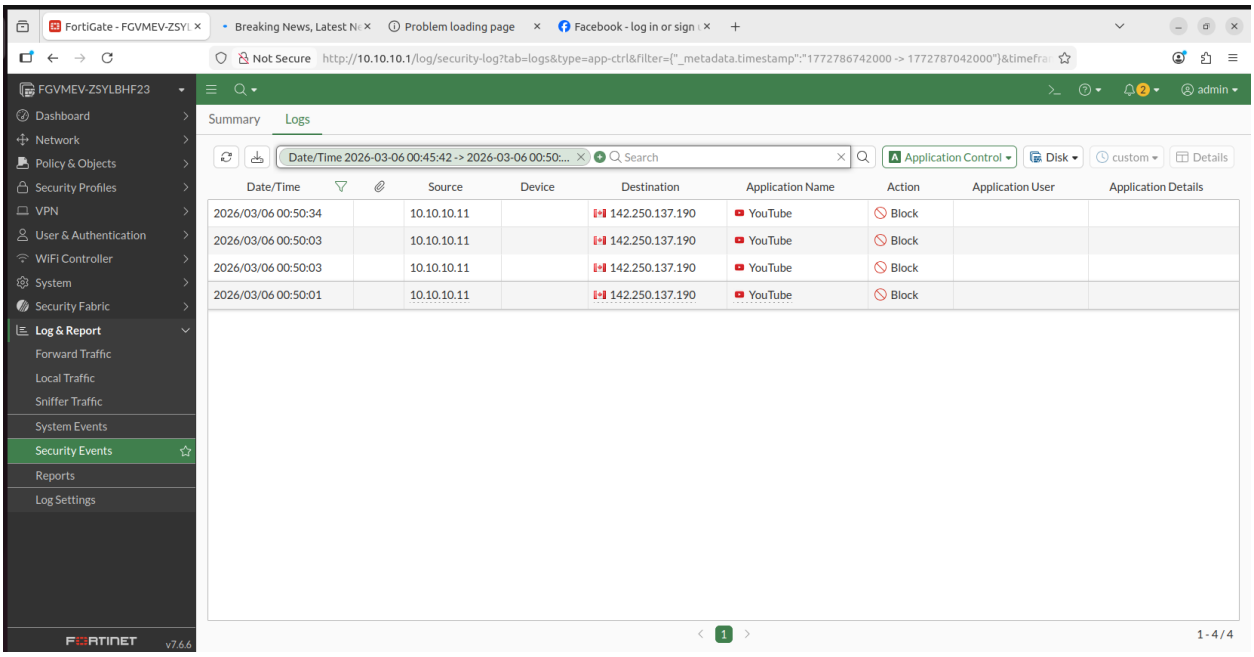


Firewall Policy:

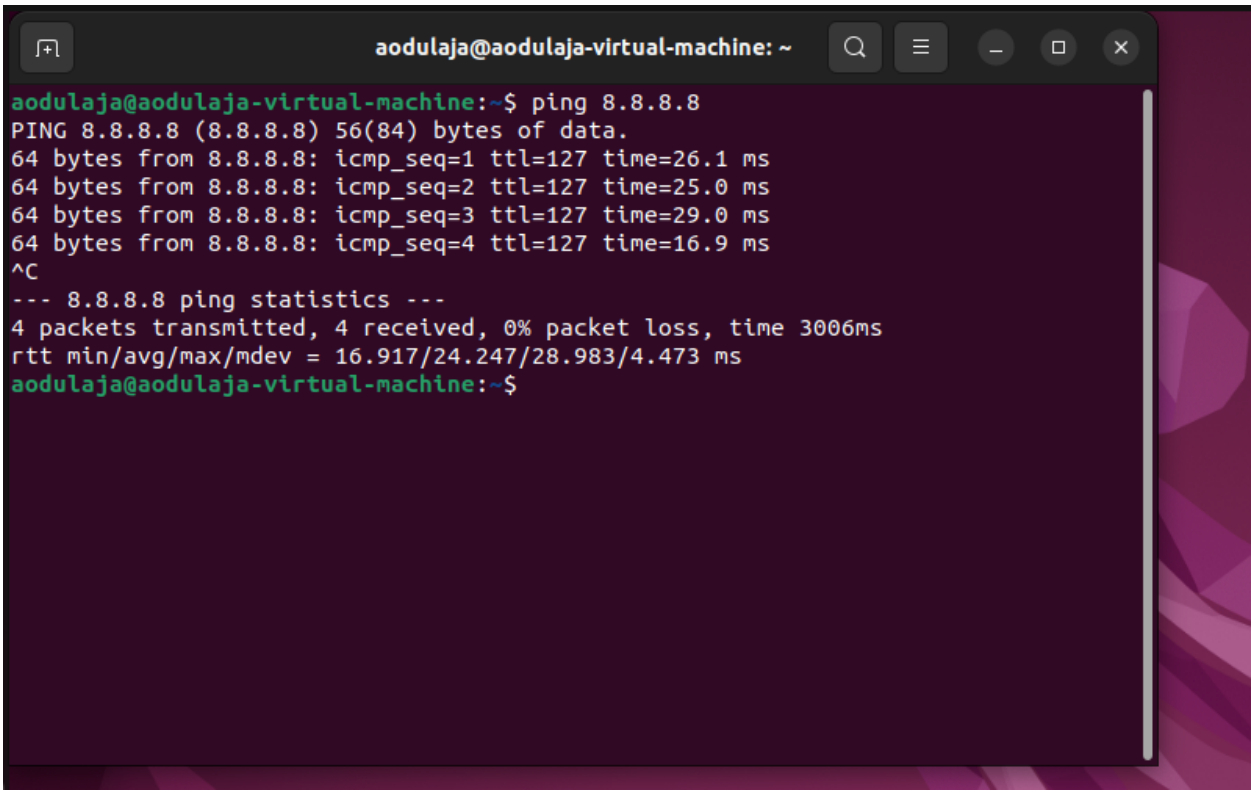




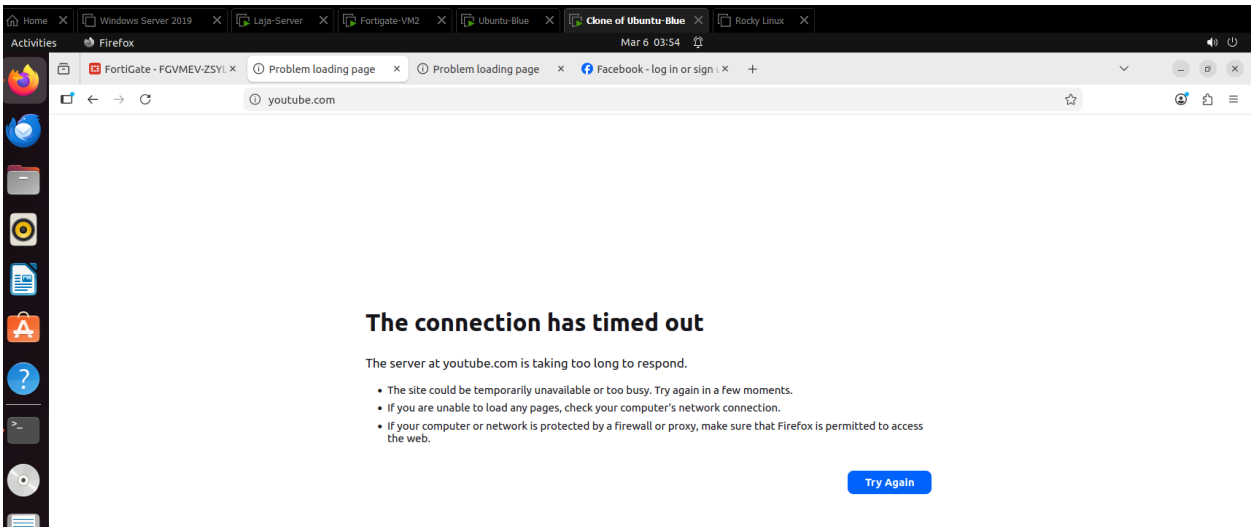
Application Control Logs:



Internet Works:



YouTube is Blocked:



10. Conclusion

This project successfully demonstrated the implementation of core firewall security controls using a FortiGate firewall in a virtual lab environment. Through the configuration of firewall policies, web filtering, and application control, the network administrator was able to regulate internet usage and enforce security restrictions on internal clients.

The exercise provided hands-on experience in traffic filtering, firewall policy creation, and security monitoring—skills that are essential in modern cybersecurity and network administration roles.

Overall, the lab illustrates how FortiGate firewalls can be used to implement layered network security controls to protect organizational networks from unwanted traffic and policy violations.